

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Case Study

COURSE NAME: COMPUTER VISION with Open CV
COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO3: *Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BTL 4)*

Assessment Tool Weightage: 20%

Read the given scenario carefully and analyze the problem using appropriate computer vision techniques. Justify your answers with relevant reasoning and comparisons wherever necessary.

1. Preprocessing and Feature Detection

A vision system processes images captured under varying lighting conditions. Without preprocessing, feature detection results are inconsistent.

Analyze the issue and explain how preprocessing improves feature detection performance.

ANS:

The main problem is that **varying lighting conditions change the pixel intensity and contrast** of the image. As a result, feature detectors may detect different or incorrect features in images of the same object.

Preprocessing improves feature detection by:

- **Normalization:** Adjusts pixel intensity to a common range, reducing the effect of brightness variations.
- **Histogram equalization/CLAHE:** Improves contrast and makes weak features more visible.
- **Noise removal:** Filters such as Gaussian or median filtering reduce unwanted variations that may create false features.
- **Grayscale conversion:** Simplifies the image while retaining intensity information needed by many feature detectors.

Comparison:

Without Preprocessing

Features vary with lighting

More false detections

Weak features may be missed

Lower detection reliability

With Preprocessing

Features are more consistent

Fewer false detections

Important features become clearer

Higher detection reliability

Conclusion:

Preprocessing reduces the influence of lighting and noise, producing more consistent image characteristics. Therefore, applying **normalization, contrast enhancement, and appropriate noise filtering before feature detection** improves the accuracy and reliability of the vision system.

2. Edge vs Corner Detection

In an image containing both straight edges and textured regions, a system uses edge detection but fails to identify key points for matching.

Analyze why edge detection is insufficient and justify the use of corner detection in this case.

ANS:

Analysis:

Edge detection identifies **boundaries where intensity changes significantly**, such as straight lines and object borders. However, an edge does not provide a unique location for matching because many points along the same edge can look similar. Therefore, using only edge detection may fail to identify distinctive key points.

Why Corner Detection is Better:

Corner detection identifies points where intensity changes significantly in **two directions simultaneously**. These points are more distinctive and stable, making them suitable for feature matching.

Comparison:

Edge Detection

Detects boundaries/lines

Information mainly along one direction

Many similar points along an edge

Less suitable for point matching

Example: Canny, Sobel

Corner Detection

Detects distinctive points

Intensity changes in multiple directions

More unique key points

Suitable for image matching

Example: Harris, Shi-Tomasi

Conclusion:

Edge detection is useful for identifying **object boundaries**, but it is insufficient for reliable point-based matching. Since the image contains textured regions and requires key points, **corner detection such as Harris Corner Detection** is more appropriate because it provides distinctive and localized features for matching and tracking.

3. Hough Transform in Noisy Images

A system uses Hough Transform to detect lines in noisy images but produces inaccurate results.

Analyze the effect of noise on detection and suggest improvements based on preprocessing or parameter selection.

ANS:

Analysis:

Noise creates unwanted intensity variations and false edges in an image. Since the **Hough Transform detects shapes by accumulating votes from edge points**, noisy edges can generate incorrect votes, resulting in false or inaccurate line detections.

Effect of Noise:

- Produces false edges.
- Creates incorrect Hough-space peaks.
- Causes detection of unwanted lines.
- May reduce the accuracy of actual line detection.

Improvements:

1. **Preprocessing:** Apply Gaussian or median filtering before edge detection to reduce noise.
2. **Canny Edge Detection:** Use suitable threshold values to obtain cleaner edges.
3. **Hough Parameters:** Adjust the accumulator threshold so that only lines receiving sufficient votes are detected.
4. **Minimum line length:** Set an appropriate minimum line length to eliminate short false lines.
5. **Maximum line gap:** Adjust the allowed gap to connect broken parts of genuine lines.

Comparison:

Without Improvement	With Improvement
Many false lines	Fewer false detections
Noisy edge map	Cleaner edge map
Low detection accuracy	Higher accuracy

Without Improvement With Improvement

Unstable results

More reliable results

Conclusion:

Noise significantly affects Hough Transform because false edges contribute unwanted votes.

Noise filtering followed by proper Canny and Hough parameter selection can substantially improve line detection accuracy and reliability.

4. Feature Matching under Transformations

Two images of the same object are taken at different scales and orientations. Traditional feature matching fails, but scale invariant feature matching works effectively.

Analyze why scale invariant methods perform better in this scenario.

ANS:

Analysis:

Traditional feature matching often depends on the direct comparison of image features. When the same object is captured at a **different scale or orientation**, the appearance and location of its features change, causing conventional methods to fail or produce incorrect matches.

Why Scale-Invariant Methods Perform Better:

Scale-invariant methods such as **SIFT (Scale-Invariant Feature Transform)** detect keypoints across multiple scales and generate descriptors based on the local structure around each keypoint. The descriptors are also designed to be relatively **rotation-invariant**, so the same feature can be recognized even when its orientation changes.

Comparison:

Traditional Matching

Sensitive to scale changes

Sensitive to rotation

Fewer reliable matches

May fail when object size changes

Suitable for similar image conditions

Scale-Invariant Matching

Handles scale changes

Handles rotation effectively

More reliable matches

Recognizes features at different sizes

Suitable for transformed images

Example:

If an object is enlarged in one image and rotated by 45° in another, traditional pixel-based or fixed-scale matching may fail because corresponding features no longer have the same appearance. **SIFT** can detect the corresponding features at different scales and orientations, resulting in more accurate matches.

Conclusion:

Scale-invariant feature methods perform better because they are designed to remain stable under **scale and orientation changes**. Therefore, algorithms such as **SIFT** are preferred for image matching, object recognition, image stitching, and tracking when images may undergo geometric transformations.

5. PCA for Feature Optimization

A system extracts a large number of features, leading to high computational cost and redundancy.

Analyze how Principal Component Analysis helps in reducing dimensionality and improving efficiency.

ANS:

Analysis:

Extracting a large number of features can create **redundant and correlated information**, increasing memory usage, processing time, and computational cost. PCA helps by transforming the original features into a smaller set of important **principal components**.

How PCA Improves Efficiency:

1. **Reduces dimensionality:** Converts many original features into fewer components.
2. **Removes redundancy:** Correlated features are represented by fewer components.
3. **Preserves important information:** Components are selected based on the amount of variance they retain.
4. **Reduces computation:** Machine-learning algorithms process fewer features, improving speed.
5. **Reduces storage:** The smaller feature representation requires less memory.

Example:

If an image produces **100 features**, PCA might reduce them to **20–30 principal components** while retaining most of the important information. The exact number depends on the selected variance threshold.

Comparison:

Without PCA	With PCA
Large feature set	Smaller feature set
More redundancy	Less redundancy
Higher computation	Lower computation
Higher memory usage	Lower memory usage

Without PCA

Longer processing time

With PCA

Faster processing

Conclusion:

PCA provides an efficient feature representation by reducing dimensionality while preserving the most significant information. Therefore, it is useful for **high-dimensional computer vision and machine-learning applications**, especially when computational efficiency and reduced redundancy are important.

Code of Conduct:

I AMALRAJ P (192524089) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided

Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand
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